Lab 04 Enterprise LAN Security Assessment

Title Page

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Lab Title: Lab 04 Enterprise LAN Security Assessment

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Introduction

This lab focused on designing and securing a segmented enterprise network that separates IT and OT networks. The lab demonstrated OSI layers, routing between networks, and secure protocol implementations such as HTTPS and SSH. The goal of the lab was to understand network architecture, protocol security, and secure remote management in enterprise environments.

Part 1 – Network Architecture & OSI Layers 1–3

Topology Description

The WBMUD network was designed using three main operational segments and one DMZ segment. The IT network represented the office environment, the OT network represented industrial control and monitoring systems, and the Remote Pump Station represented a remote operational site connected through WAN links. A DMZ attacker node was also included for security testing. Each network was assigned its required IP range, and routers were connected using serial links to support communication between segments.

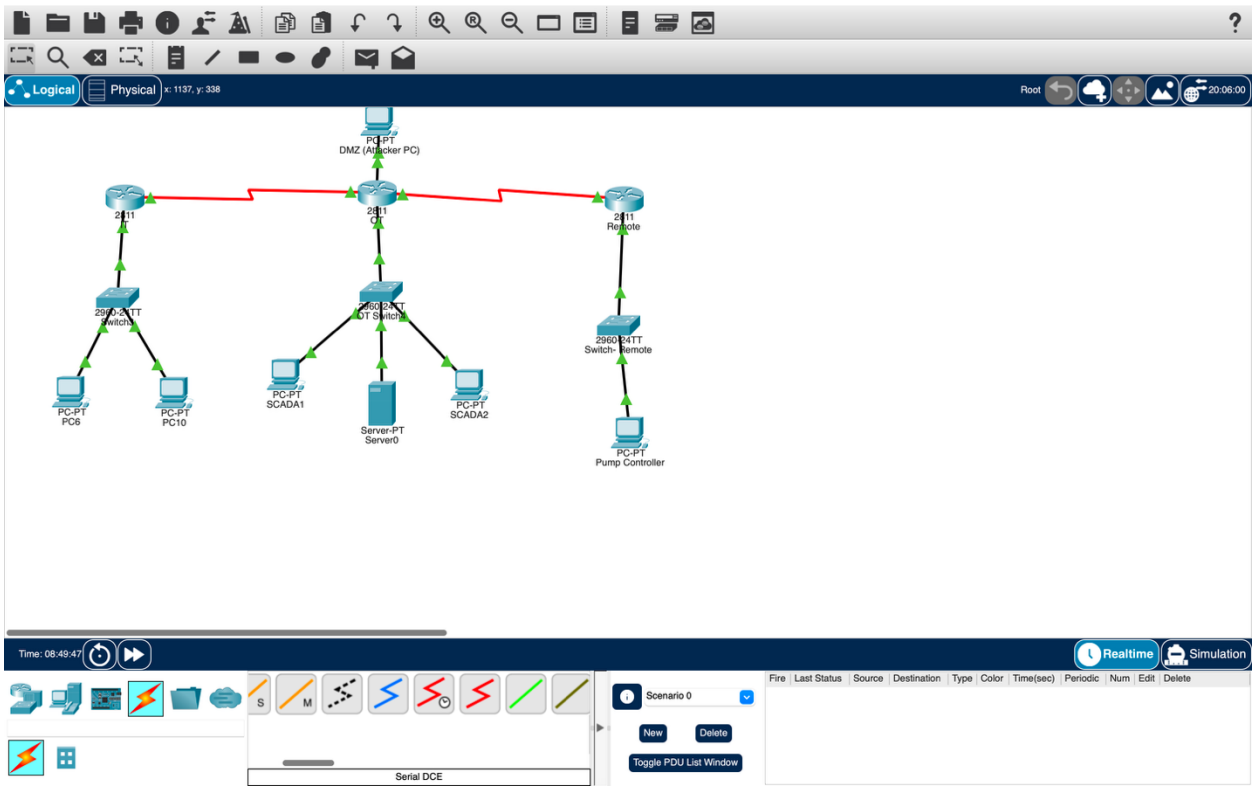

Figure 1. Full topology view.

OSI Layer 1 Reflection

The Physical layer is responsible for transmitting raw bits over a physical medium. In this lab, cables, ports, and powered devices represent Layer 1 because they provide the physical connection required for communication. This layer does not handle IP or MAC addressing. It focuses only on the transmission of bits between devices.

OSI Layer 2 Reflection

OSI Layer 2 is the Data Link layer, responsible for MAC addresses and switching. Switches forward frames based on MAC addresses. The MAC address table shows how switches learn device addresses and forward traffic within the local network.

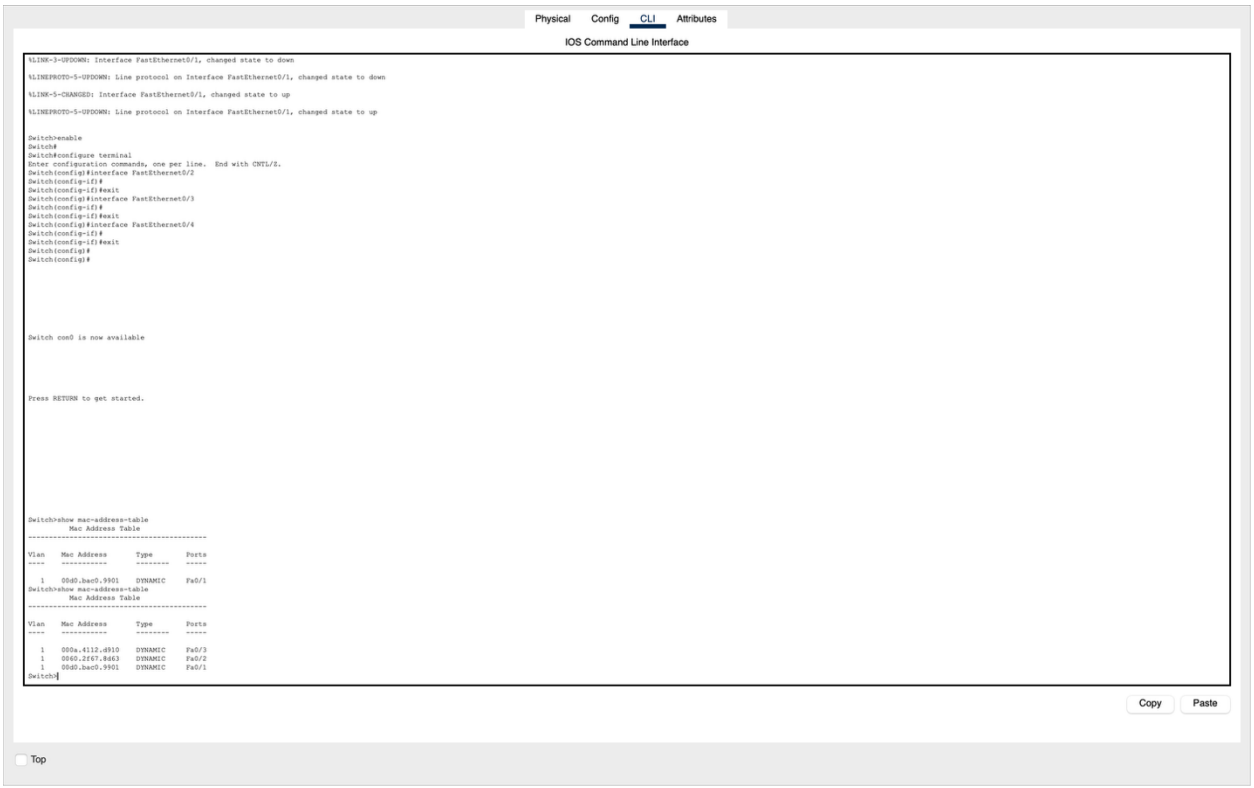

Figure 2. Switch MAC address table showing dynamically learned MAC addresses and associated switch ports.

OSI Layer 3 Reflection

OSI Layer 3 is the Network layer, responsible for IP addressing and routing. Routers use routing tables to forward packets between different networks. In this lab, routers allowed communication between IT, OT, and Remote networks using IP routing.

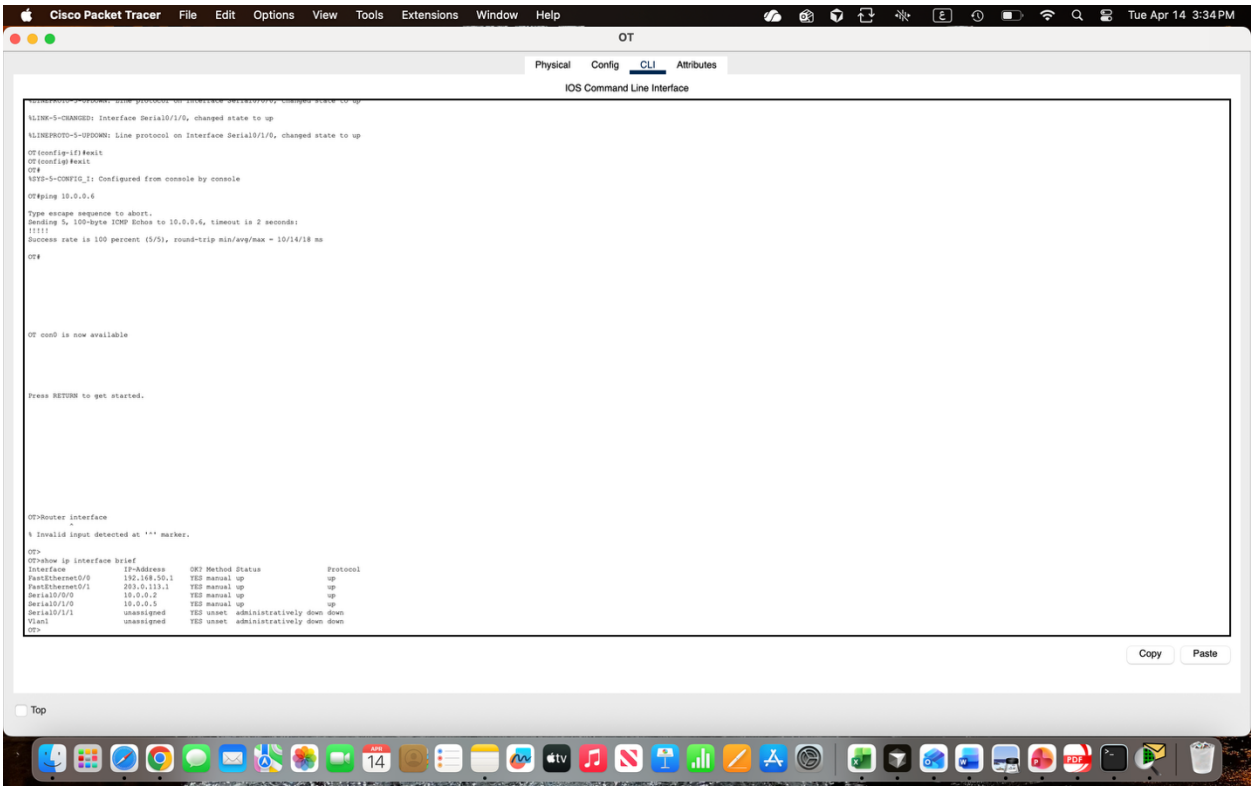

Figure 3. Router interface summary showing assigned IP addresses and operational interface status.

OSI Layer 3 Reflection

The Network layer is responsible for logical addressing and routing between different networks. In this lab, IP addressing, subnet masks, default gateways, and static routes were all configured at Layer 3. Routers used this information to forward packets between the IT, OT, and Remote Pump Station networks. Without Layer 3 routing, devices would only be able to communicate within their own subnet.

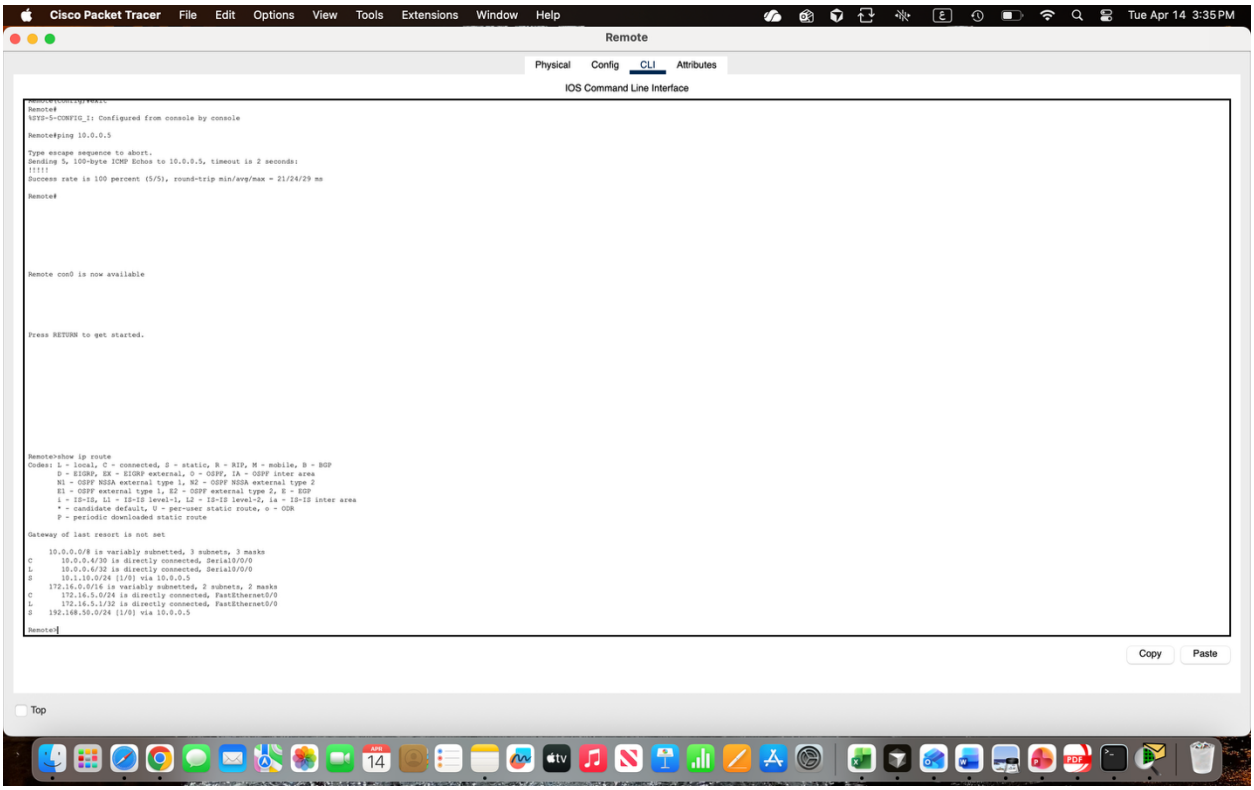

Figure 4. Routing table showing directly connected networks and configured static routes between WBMUD segments.

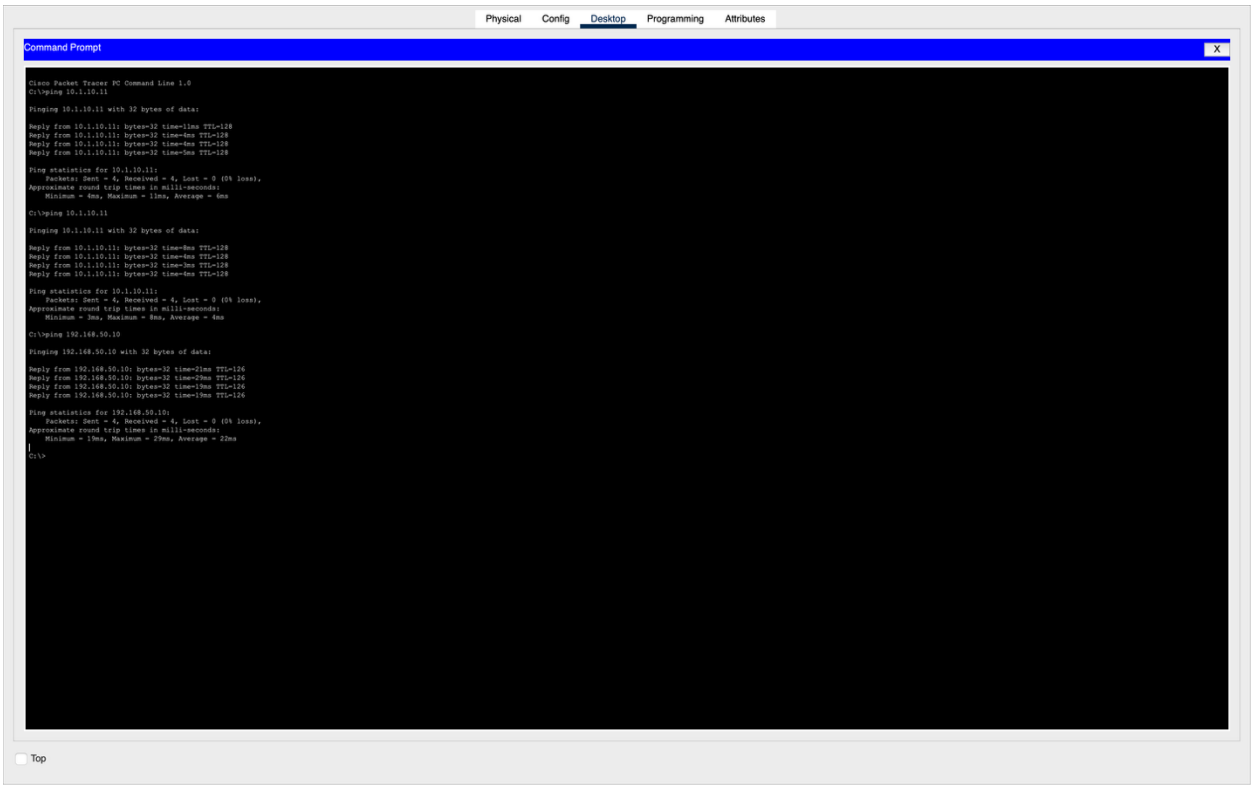

Figure 5. Connectivity verification from the IT workstation showing successful pings to a local IT host, the OT web server, and the Remote Pump Station controller.

Troubleshooting Notes

Diagnosis:

Initial connectivity problems were caused by incorrect IP addressing and incomplete routing verification. In the earlier version of the topology, the OT web server was configured with an incorrect IP address, and some required routing evidence was not included in the report.

Resolution:

The OT web server IP address was corrected to 192.168.50.10, and router interface and routing table verification were completed using show ip interface brief and show ip route. After correcting the addressing and confirming static routes, all required ping tests from the IT workstation were successful.

Part 2 – Protocol Security & OSI Layers 4–7

This section analyzes the security implications of TCP vs UDP, HTTP vs HTTPS, and SSH vs Telnet in the WBMUD network.

TCP vs UDP Security Explanation

TCP provides greater security visibility because it uses a three-way handshake: SYN, SYN-ACK, and ACK. This makes TCP communication stateful, which allows firewalls and intrusion detection systems to track sessions more effectively. UDP, by contrast, is connectionless and sends datagrams without establishing a session first. As a result, UDP offers less visibility and less reliable state tracking for security monitoring.

OSI Layer 4 Reflection

TCP and UDP operate at OSI Layer 4, the Transport layer. TCP provides reliable communication using acknowledgments, sequencing, and connection establishment, but it is slower. UDP is faster because it does not establish a connection or guarantee delivery, but it is less reliable. From a security perspective, TCP is easier for firewalls and IDS/IPS systems to track because it is stateful, while UDP is stateless and harder to monitor.

HTTP vs HTTPS Security

HTTPS is more secure than HTTP because it uses TLS/SSL encryption to protect data in transit between the client and the server. HTTPS provides encryption, integrity protection, and server authentication using digital certificates, which helps prevent man-in-the-middle attacks. HTTP sends data in clear text, making it easy for attackers to intercept or modify the data.

Figure 6. Browser access to the OT web server using both HTTP and HTTPS protocols.

OSI Layer Mapping for Encryption

TLS/SSL encryption used in HTTPS is generally considered part of the Application layer (Layer 7) of the OSI model, sometimes described across Layers 5–7. TLS operates on top of TCP and encrypts application data before it is transmitted over the network. This protects the data while it is transported by TCP at the Transport layer.

SSH vs Telnet Security

SSH replaced Telnet because Telnet sends usernames, passwords, and commands in clear text, which can be intercepted by attackers. SSH encrypts the entire remote management session, including login credentials and commands, providing confidentiality and integrity. This makes SSH much more secure and protects against eavesdropping and session hijacking. I used a serial DCE cable.

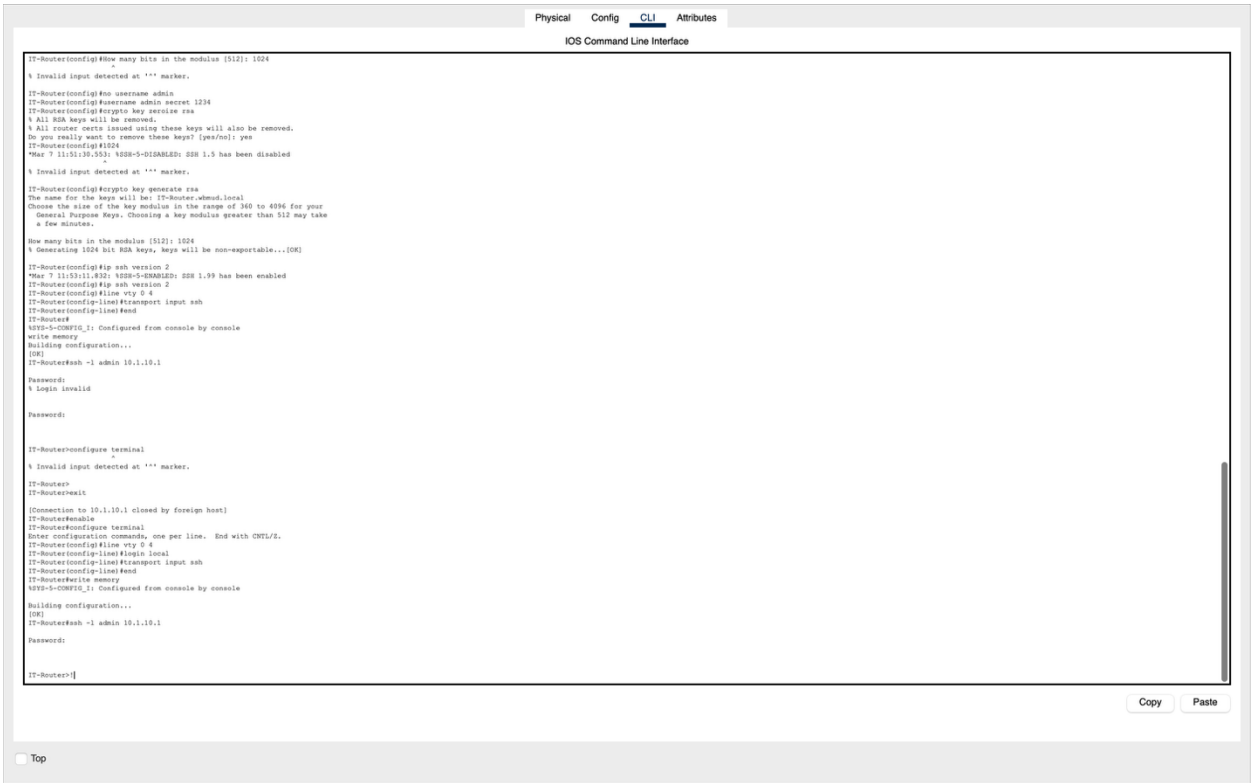

Figure 7. Successful SSH session from the IT workstation to the IT router, demonstrating encrypted remote management access.

DMZ Attacker Node

The DMZ Attacker PC was added to simulate an external threat to the network. This device represents a potential attacker located outside the internal IT and OT networks. It was configured with the IP address **203.0.113.100** and connected directly to the router to create a separate DMZ segment.

The purpose of this setup is to test network security and evaluate how the system responds to external access attempts. By isolating the attacker in a DMZ, the network design ensures that internal systems remain protected while still allowing controlled testing.

A successful ping test between the attacker PC and the router confirmed proper connectivity and correct configuration of the DMZ network.

Building on the previous network design and protocol analysis, the following section explores a real-world vulnerability scenario and incident response within the WBMUD network.

Part 3 – Shellshock Vulnerability Analysis

Shellshock is a critical vulnerability found in the Bash shell, which is commonly used in Linux and Unix-based systems. This vulnerability allows attackers to execute malicious commands remotely by exploiting how Bash processes environment variables.

In a web server environment, Shellshock can be triggered through CGI (Common Gateway Interface) scripts. An attacker can send specially crafted HTTP requests that include malicious code in headers such as User-Agent. When the server processes the request, Bash unintentionally executes the injected commands.

In the WBMUD network scenario, the attacker node in the DMZ could exploit this vulnerability to gain unauthorized access to the OT web server. Once inside, the attacker may execute commands, access sensitive data, or move laterally within the network.

This vulnerability highlights the importance of keeping systems updated and properly patched. Without patching, even a simple web server can become an entry point for a major network compromise.

Part 4 – Incident Response (Attack Path)

The attack begins from the DMZ attacker node, which represents an external threat. The attacker sends a malicious request to the OT web server by exploiting a vulnerability such as Shellshock.

Once the OT server is compromised, the attacker gains access to the internal OT network. From there, the attacker may attempt to move laterally toward other systems, including SCADA devices or even the IT network through routing connections.

This demonstrates how a single vulnerable system in the OT environment can expose the entire enterprise network.

Part 4 – Root Cause Analysis

The primary cause of the attack is the presence of an unpatched vulnerability in the web server. The system failed to apply security updates, leaving it exposed to known exploits such as Shellshock.
Another contributing factor is the lack of proper access control and network segmentation enforcement. Although the network is segmented, insufficient security controls allow the attacker to interact with the OT server.

Additionally, the absence of monitoring and intrusion detection systems makes it difficult to detect and stop malicious activity in real time.

Part 4 – Remediation Plan

To prevent similar attacks in the future, several security measures should be implemented.

First, all systems must be regularly updated and patched to eliminate known vulnerabilities such as Shellshock. Second, a firewall should be configured to restrict access between network segments and block unauthorized traffic.

Third, secure communication protocols must be enforced. For example, HTTPS should replace HTTP, and SSH should be used instead of Telnet for remote access.

Additionally, intrusion detection and monitoring systems should be deployed to identify suspicious behavior. Finally, strong access control policies should be applied to limit user privileges and reduce the risk of unauthorized access.

Part 5 – OSI Model Summary

Layer 1 (Physical): Cables and device connections
Layer 2 (Data Link): MAC addresses and switch forwarding
Layer 3 (Network): IP addressing and static routing
Layer 4 (Transport): TCP vs UDP communication
Layer 7 (Application): HTTP, HTTPS, SSH protocols

Reflection

In this lab, I learned how network design and security are closely connected, especially in critical infrastructure environments like water and wastewater systems. Building the WBMUD network helped me understand the importance of proper segmentation between IT, OT, and remote networks to reduce security risks.

One of the key lessons from this lab is that even a well-designed network can become vulnerable if security practices are not properly implemented. For example, the Shellshock vulnerability showed how a single unpatched system can allow attackers to gain access and potentially compromise the entire network.

I also learned the importance of secure communication protocols. Using HTTPS instead of HTTP and SSH instead of Telnet significantly improves the security of data transmission and remote management. These protocols help protect sensitive information from being exposed to attackers.

Another important takeaway is the role of monitoring and incident response. Identifying how an attack occurs and understanding its root cause are essential steps in preventing future incidents. Implementing security controls such as firewalls, intrusion detection systems, and access restrictions can greatly enhance the overall security posture of the network.

Overall, this lab provided a practical understanding of how networking concepts and cybersecurity principles work together. It also showed me the importance of applying both technical skills and security awareness when designing and managing enterprise networks.

Conclusion

This lab demonstrated how enterprise networks use segmentation and secure protocols to protect critical infrastructure systems. The lab showed the importance of using secure protocols such as HTTPS and SSH instead of HTTP and Telnet. The key takeaway is that protocol security and OSI layer understanding are essential for designing secure enterprise IT and OT networks

GitHub Evidence Locker
The supporting evidence for this lab, including screenshots and documentation, is stored in the following GitHub repository:
https://github.com/N-alaiban/Lab-4-Enterprise-LAN-Security.md